

Thompson Wong

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EDUCATION

University of California, San Diego

La Jolla, CA

B.S. in Mechanical Engineering — Specialization in Controls and Robotics

Expected June 2027

- **Honors:** EMPOWER Scholar (Awarded by National Science Foundation) **GPA: 3.8/4.0**
- **Courses:** Engineering Graphics & Design, Properties of Materials, Elements of Materials Science, Fluid Mechanics, Thermodynamics, Solid Mechanics, Linear Circuits, Statics, Dynamics, Computational Methods in Engineering

Foothill College

Los Altos Hills, CA

A.S. in Mechanical Engineering - Dean's List - GPA: 3.72/4.0

Fall 2022 - Spring 2024

TECHNICAL SKILLS

Computer Aided Design: SolidWorks, OnShape, AutoDesk Fusion (Rapid Prototyping, Design for Manufacturing)

Manufacturing: Manual & CNC Mill, Lathe, 3D-Printer (FDM & SLA), Laser-Cutter, Hand & Power Tools, Soldering

Software: Proficiency in MATLAB, Simulink, Java, C++, LaTeX, Microsoft Suite, Google Suite, Adobe Creative Suite

EXPERIENCE

Research Intern - Morimoto's Lab (Surgical Robotics Technology)

June 2025 - Present

Contextual Robotics Institute (Medical and Healthcare Robotics)

La Jolla, CA

- Reserached & developed a customized end effector for bronchoscopy (Bidirectional Notched Nitinol Tube Wrist)
- Characterized design geometric parameters experimentally to enable repeatable and robust bending and actuation
- Performed FEA simulations to evaluate strain limit & structural integrity under actuation loads to iterative design
- Developed customized fixture design and CAM settings for micro-milling 3.6mm outer diameter nitinol tubing

Chassis Engineer - UCSD Formula SAE (Triton Racing)

October 2024 - Present

- Designed chassis frame welding jig with calculated tolerances & geometry to reduce thermal warping during weld
- Optimized frame's stiffness-to-weight ratio by simulating torsional stiffness via Finite Element Analysis (FEA)
- Developed frame's shear panels using Sheet Metal Design, and documented its design and manufacturing process
- Created detailed engineering drawings and bill-of-materials (BOM) for multiple ten-plus-part assemblies

Undergraduate Researcher - Delson's Lab (Medical Device Technology)

January - June 2025

Center for Memory and Recording Research, UC San Diego Health Science

La Jolla, CA

- Reserached & developed an ergonomic endoscopic gripper to reduce physician muscle strain during procedures
- Designed iteratively to optimize tool's grip angle and internal springs to enhance torque and minimize slippage
- Utilized surface electromyography (sEMG) sensors and testing-apparatus to quantify muscle intensity & fatigue

Project Manager - American Society of Mechanical Engineers @UCSD

July 2025 - Present

- Represented UCSD at professional board meetings to coordinate resources and opportunities for student members
- Oversaw budget, scheduling, and progress of two hands-on projects while mentoring members in technical skills

President - Engineering Club @Foothill College

September 2023 - July 2024

- Worked 5 hrs/week to coordinate weekly meetings and maintained regular communication with faculty advisors
- Managed project workflows with Jira and Notion, overseeing progress on three concurrent technical projects

PROJECTS

6DOF Chess Cobot (Universal Robots UR10) | Club Project

September 2023 - July 2024

- Designed and assembled a mobile base for securing arm to engage in a game of chess against a human player
- Collaborated with the software team to determine optimal camera placement for computer vision performance
- Developed an electromagnetic gripper module compatible with the UR10 for maneuvering chess pieces

Quadruped Robot Dog | Stanford University (CS123) - Independent Study

March - September 2024

- Programmed 6-DOF legs to move using inverse kinematics with Proportional-Integral-Derivative feedback control
- Created a cartesian-space safety limit for each joint utilizing forward kinematics with impedance control
- Utilized GPU-based parallel-simulation as a reinforcement learning platform to optimize puppy's walk speed